

**SECTION A**Answer **all** questions.

1. Addition of aqueous lead nitrate to aqueous potassium iodide causes a precipitate to form. Write an ionic equation for the formation of the precipitate, including state symbols. [1]

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2. Use the data shown to explain why silver chloride is insoluble in water. [2]

|                                                  | Enthalpy / $\text{kJ mol}^{-1}$ |
|--------------------------------------------------|---------------------------------|
| Standard enthalpy of hydration for $\text{Ag}^+$ | -464                            |
| Standard enthalpy of hydration for $\text{Cl}^-$ | -364                            |
| Enthalpy of lattice formation of $\text{AgCl}$   | -905                            |

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3. A white crystalline solid gives a red colour in a flame test and addition of concentrated sulfuric acid to the solid gives purple fumes and a smell of rotten eggs. Name the ions present, giving your reasons. [2]

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4. Aluminium chloride forms  $\text{Al}_2\text{Cl}_6$  dimers using coordinate bonding. Explain why the coordinate bonds form. [2]

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5. Addition of ammonia solution to aqueous copper(II) sulfate causes a pale blue precipitate to form followed by a coloured solution when excess ammonia solution is added.

Give the colour of the solution formed.

[1]

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6. The Haber process uses a heterogeneous catalyst.  
State what is meant by the term 'heterogeneous' in this context.

[1]

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7. Hexagonal boron nitride is sometimes called white graphite. Give **one** difference between the physical properties of hexagonal boron nitride and graphite.

[1]

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